Caterina Beatrice Leimer Saglio

PhD Candidate in Mathematical Models and Methods in Engineering
Politecnico di Milano - MOX Laboratory, Italy

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WORK EXPERIENCE

PhD Candidate in Mathematical Models and Methods in Engineering *Sep 2023 - Present*
Politecnico di Milano - MOX Laboratory, Italy

Thesis: Computational modeling of the human brain: from physiology to neurodegenerative diseases.
Advisors: Prof. P. F. Antonietti, Prof. S. Pagani

INVITED RESEARCH VISIT

Visiting PhD Student *Jan 2026 - April 2026*
Simula Research Laboratory - Oslo, Norway *Jan 2025 - April 2025*
Numerical Analysis and Scientific Computing department

Invited by: Prof. K. A. Mardal

EDUCATION

MCs in Mathematical Engineering, Computational Science major *Sep 2020 - May 2023*
Politecnico di Milano, Italy

Master thesis: An adaptive discontinuous Galerkin method for the numerical modeling of epileptic seizures.
Advisors: Prof. P. F. Antonietti, Prof. S. Pagani, Dr. M. Corti

Erasmus+ Experience *Sep 2021 - Feb 2022*
Bordeaux INP, ENSEIRB-MATMECA, Bordeaux, France.

Relevant Subjects : Computational Fluid Dynamics, Fluid Labs.

BCs in Mathematical Engineering *Sep 2017 - Sep 2020*
Politecnico di Milano, Italy

PUBLICATIONS

Peer-Reviewed Journal Papers

(J1) **Leimer Saglio, C.B.**, Pagani, S., Corti, M. and Antonietti, P. F. (2026). *A high-order Discontinuous Galerkin method for the numerical modeling of epileptic seizures*. Computers & Mathematics with Applications. 205, 112-131.

(J2) **Leimer Saglio, C.B.**, Pagani, S. Antonietti, P. F. (2025). *A p-adaptive polytopal discontinuous Galerkin method for high-order approximation of brain electrophysiology*. Computer Methods in Applied Mechanics and Engineering, 446, 118249.

Accepted Papers

(A1) **Leimer Saglio, C.B.**, Pagani, S. and Antonietti, P. F. *A massively parallel non-overlapping Schwarz* *preconditioner for PolyDG methods in brain electrophysiology* - arXiv:2512.19536 (2025)

Submitted Papers

(S1) Antonietti, P. F., Corti, M., Pagani, S. and **Leimer Saglio, C.B.** *The `lymph 2.0` library: p-adaptive algorithms and parallel assembly strategies for polytopal DG methods* - arXiv:2606.24646 (2026)

(S2) **Leimer Saglio, C.B.**, Corti, M., Pagani, S. and Antonietti, P. F. *A novel mathematical and computational framework of amyloid-beta triggered seizure dynamics in Alzheimer's disease* - arXiv:2511.10369 (2025)

(S3) Daniele, F., **Leimer Saglio, C.B.**, Pagani, S. and Antonietti, P. F. *Mathematical and numerical modeling of coupled oxygen dynamics and neuronal electrophysiology* - arXiv:2509.22863 (2025)

Theses

(T1) *An adaptive discontinuous Galerkin method for the numerical modeling of epileptic seizures* – Master Thesis in Mathematical Engineering: Computational Science in Engineering (2023).

TEACHING ACTIVITIES

Teaching Assistant : Curve e superfici per il design — *A.Y. 2025/26*
MSc in Fashion Design. Politecnico di Milano.
Lecturer: A. Scotti

Tutor: Numerical Analysis for Partial Differential Equations — *A.Y. 2023/24*
MSc in Mathematical Engineering.Politecnico di Milano. — *A.Y. 2024/25*
Lecturer: P.F.Antonietti

Tutor: Scientific Computing Tools for Advanced Mathematical Modeling — *A.Y. 2023/24*
MSc in Mathematical Engineering.Politecnico di Milano. — *A.Y. 2024/25*
Lecturer: S.Pagani

INVITED LECTURES, CONFERENCE TALKS, SEMINARS

Invited Talks at Conferences

WCCM–ECCOMAS 2026	Contributed talk within the minisymposium "Computational Brain Multiphysics" *An adaptive high-order polytopal method for coupled neuronal electrophysiology and amyloid-beta spreading* **Organizer:** I. Fumagalli **Dates and Location:** July 19–24, 2026 – Munich, Germany
LYNUM VIII	Invited talk at the Lombardy Young NUMerical Analysts Workshop *Modelling neuronal electrophysiology with high-order polytopal methods* **Organizers:** K. B. Haile and M. Mosconi **Date and Location:** June 5, 2026 – Milano, Italy

GIMC SIMAI YOUNG 2026	Contributed talk within the thematic session ”High-Order Numerical Methods for Complex Mechanics and Higher-Order PDEs” *A p-adaptive high-order polytopal method for modelling neuronal electrophysiology* **Organizers:** I. Bioli and L. Greco Contributed talk within the thematic session ‘Robust Preconditioning Techniques for Scientific Applications *A non-overlapping Schwarz preconditioner for PolyDG methods in brain electrophysiology* **Organizers:** M. Feder, F. Mugnaioni, and D. Polverino **Dates and Location:** June 3–5, 2026 – Pisa, Italy
SIAM CHAPTERS MEETING 2026	Invited talk within the thematic session ”Models and Methods for Healthcare Applications” **Organizers:** S. A. Gomez and I. Fumagalli **Dates and Location:** January 23-30, 2026 - Milano (Italy)
YIC 2025	Invited talk within the minisymposium ”Recent Advances in Polytopal Finite Element Methods” **Organizers:** S. A. Gomez and I. Fumagalli **Dates and Location:** September 17-19, 2025 - Pescara (Italy)
ENUMATH 2025	Invited talk within the minisymposium ”Multiphysics Computational Modeling of the Brain Function” **Organizers:** M. Rognes, P. F. Antonietti, M. Kutcha and I. Fumagalli **Dates and Location:** September 1-5, 2025 - Heidelberg (Germany)
DOMAIN DECOMPOSITION METHODS 2025	Invited talk within the minisymposium ”Next-geneneration numerical methods for computational life sciences” **Organizers:** F. Bonizzoni, S. Pagani and M. Corti **Dates and Location:** June 23-27, 2025 - Milan (Italy)
COUPLED PROBLEMS 2025	Invited talk within the minisymposium ”High-Order and Innovative Methods for Coupled Problems in Life Science and Geophysics” **Organizers:** F. Bonizzoni, S. Pagani and M. Corti **Dates and Location:** May 25-28, 2025 - Villasimius (Italy)
GIMC SIMAI YOUNG 2024	Invited talk within the minisymposium ”Advanced numerical methods for coupled problems on complex domains” **Organizers:**A. Borio, F. Marcon and M. Visinoni **Dates and Location:** July 10-12, 2024 - Naples (Italy)
ECCOMAS 2024	Invited talk within the minisymposium ”Numerical Methods for the Multiphysics Modeling of Brain Function” **Organizers:** P. Antonietti, F. Bonizzoni, I. Fumagalli and K. A. Mardal **Dates and Location:** June 3-7, 2024 - Lisbon (Portugal)

Contributed Talks

SOFTMECH 2025	Presentation: ”A numerical study of the eletrophysiological substrate of epilepsy ” **Dates and Location:** June 11-13, 2025 - Milano (Italy)
THE NUMERICAL BRAIN 2024	Lightning presentation: ”Numerical modeling of neuronal electrophysiology” **Dates and Location:** October 21-23, 2024 - Amsterdam (Netherlands)

Poster sessions

- NEMESIS Kick-off workshop *June 2024*
  Dates and Location: June 19-21, 2024 - Montpellier (France)

- Mathematics for our Health (M4H) *Nov 2024*
  Dates and Location: Nov 7-8, 2024 - Milan (Italy)

SUMMER SCHOOLS, PhD COURSES

Summer Schools

- Isogeometric Analysis: Theory, Applications, and New Trends *June 2024*
  Organizers: Giannelli C., Montardini M., Sangalli G.

- Course on Advanced Numerical Methods for Hyperbolic Equations *Jan 2024*
  Organizers: Dumbser M.

PhD Courses

- Fluid and flow structure interactions.
  Lecturer: Pata V. and Webster J.
- Mathematical aspects of fluid mechanics.
  Lecturer: Giorgini A.
- Research skills.
  Lecturer: Biscari P.
- Principle of computational neuroscience.
  Lecturer: Linaro D.
- Communication in science.
  Lecturer: Paganoni A.
- Mathematical and numerical foundations of scientific machine learning.
  Lecturers: Miglio E. and Pagani S.

RELEVANT PROJECTS

An adaptive discontinuous Galerkin method for the numerical modeling of epileptic seizures *Sept 2022 - May 2023*

Politecnico di Milano, [Computational Science,C++,Matlab] - Master Thesis
- Formulation of an p-adaptive discontinuous Galerkin method for the modelization of epileptic seizures.
- Modeling epileptic events in different brain tissues exploiting coupled PDEs and ODEs.

Scientific Computing Tools for Advanced Mathematical Modeling Projects *Sept 2022 - May 2023*

Politecnico di Milano, [Computational Science,C++,Python,OpenFOAM]
- Identify an optimal anti-tachycardia pacing protocol given a single heartbeat.
- Shape modeling and 2D flow solvers for left atrial appendage and analysis of risk for ischemic stroke.
- Analysis of the shape and performances of the new Formula 1 cars exploiting streamlines.

CFD simulation of the turbulence of a corium bath *Oct 2021 - Jan 2022*

Bordeaux INP, [OpenFOAM, Fluid Dynamics]
- Industrial project in collaboration with CEA industry.
- Modelling and analysis of turbulent flows to cool the recuperator.

Modeling and generation of Purkinje network in cardiac electro-physiology *Sept 2022 - May 2023*

Politecnico di Milano, [Computational Science,C++,Electrophysiology]

- Implementation of structures for the LifeX repository to handle 1D-3D problems.
- Implementation of the Eikonal model for the Purkinje network into the LifeX repository.

MENTORING ACTIVITIES

Master Theses

Citterio L.*"A p-adaptive polytopal method for modeling neurodegenerative diseases"* *A.Y. 2024/25*
Advisor: P. F. Antonietti **Co-Advisor:** M. Corti, C. B. Leimer Saglio

Daniele F.*"Analysis of interaction of oxygen dynamics with neuronal electrophysiology"* *A.Y. 2024/25*
Advisor: P. F. Antonietti **Co-Advisor:** S. Pagani, C. B. Leimer Saglio

Mancini A.*"Coupling neuronal electrophysiology with pathological β-amyloid spreads"* *A.Y. 2023/24*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:**M. Corti, C. B. Leimer Saglio

Master Projects: Course of Numerical analysis for partial differential equations

Singorelli N., Negroni A. - *"Enhancing p-Adaptive strategies with machine learning for efficient multiscale simulation"* *A.Y. 2024/25*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Ubezio E., Uboldi F. - *"Computational modeling of neural dynamics"* *A.Y. 2024/25*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Derme F., Fumagalli P. - *"hp-Adaptive strategies for high-fidelity multi-physics simulations"* *A.Y. 2024/25*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Ruggeri E., Zois S., Visalli F. - *"1D Directed Graph for efficient representation of epileptic seizures"* *A.Y. 2023/24*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Minichelli N., Polizzi M.- *"A high-order space-time Runge-Kutta DG discretization for the numerical modeling of epileptic seizures"* *A.Y. 2023/24*
Advisor: P. F. Antonietti, S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Master Projects: Course of Scientific Computing Tools for Advanced Mathematical Modeling

Vanini T., Volonteri T., Amadini A. - *"Multicompartimental neuronal model induced by calcium imbalances"* *A.Y. 2024/25*
Advisor: S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Di Noto S., Pasquer R., Muttalammashettahalli D. - *"Description of long term connections at brain tissue-scale"* *A.Y. 2024/25*
Advisor: S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Daniele F., Campioni M., Caon B. - *"Long-term connections for modeling anisotropic brain tissue"* *A.Y. 2024/25*
Advisor: S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Tegon A., Breda L., Ferrari Dagrada T - *"Modeling pre-synaptic neuronal transmission in pathological context"* *A.Y. 2023/24*
Advisor: S. Pagani **Co-Advisor:** C. B. Leimer Saglio

Casella G., Caldarini S, and Irali E. - *"Modeling activation times for pathological seizures in brain tissues"* *A.Y. 2023/24*
Advisor: S. Pagani **Co-Advisor:** C. B. Leimer Saglio

LANGUAGES

- **Italian**: Mothertongue.
- **English**: Advanced, reading and speaking. TOEIC Certificate (2020) 920/990 [B2].
- **French**: Intermediate, reading and speaking [B1].